Style & Formatting Guide

Independent Study Write-Up — Multivariate Calculus (RVCC MAT 251 / Calc III)

This guide codifies the conventions used in the `amsart` source for the MAT 251 write-up, so that every unit, problem, and proof is typeset consistently and new material can be added without re-deriving the formatting. It is organized to mirror the preamble: global layout, the package stack, the custom notation macros, structural environments, theorem environments, and the document's unit architecture. A pre-final checklist and a quick-reference cheat sheet are at the end.

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1. Document Class & Global Layout

Setting	Value	Notes
Class	`amsart`	AMS article class — provides the section styling, theorem support, and the centered `\maketitle` block used here.
Base font size	`9pt`	Set via the class option `[9pt]`. Small for body text — see §8.
Paper	`a4paper`	
Text block	`6in × 8in`	From `\usepackage[a4paper, total={6in, 8in}]{geometry}`; `total` fixes the text-area size and centers the margins.

Convention: keep all layout changes inside the `geometry` call. Don't hand-tune `\hoffset` / `\voffset`.

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2. Package Stack

Packages are grouped by purpose. Three ordering rules are load-bearing and must be preserved:

- `amsmath` / `mathtools` **before** `amsthm`
- `csquotes` **before** `biblatex`
- `hyperref` **late** (after most packages; bibliography after it)

Group	Packages	Purpose / notes
Encoding & language	`inputenc` (utf8), `babel` (english)	UTF-8 source + English hyphenation. With modern TeX Live, `inputenc` is optional but harmless.
Mathematics	`amssymb`, `mathtools` (pulls in `amsmath`), `amsthm`	Symbols, math layout, theorem environments. Order as above.
Graphics, floats & captions	`graphicx`, `float` (`[H]`), `wrapfig`, `caption`, `subcaption`, `pdfpages`	`subcaption` already loads `caption`, so the explicit `caption` line is redundant but harmless. `float` provides `[H]`; `pdfpages` enables `\includepdf`.
Tables, lists & color	`tabularx`, `enumitem`, `xcolor`	`tabularx` for width-controlled tables; `enumitem` powers the `parts` list; `xcolor` drives caption colors.
Plotting	`pgfplots` (`compat=newest`)	Native TikZ/pgfplots figures.
Boxes, spacing & misc	`tcolorbox`, `setspace`, `textcomp`, `footmisc` (`[symbol]`), `blindtext`	`tcolorbox` for framed callouts; `setspace` for `MLAENV`; `footmisc[symbol]` + the footnote redefinition give symbol footnotes; **`blindtext` is placeholder-only — remove before final.**
Page geometry	`geometry`	See §1.
Hyperlinks	`url`, `hyperref`	`url` is redundant once `hyperref` is loaded; safe to drop.
Bibliography	`csquotes` (autostyle), `biblatex` (`backend=biber, sorting=none`)	`sorting=none` ⇒ references appear in **citation order**. Source file: `MyBib.bib`. Put `\printbibliography` in the body where references should appear.

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3. Notation Macros (Math)

Use these instead of raw markup — it keeps notation uniform across units. Several macros **auto-bold** their arguments (`\proj`, `\comp`, `\vec`), so pass plain letters, not `\mathbf{...}`.

Macro	Example input	Renders as	Notes
`\anglevector{...}`	`\anglevector{1,2,3}`	⟨1, 2, 3⟩	Auto-sized angle brackets for component vectors.
`\norm{...}`	`\norm{\vec v}`	‖**v**‖	Built from doubled single bars `\left
`\unitvector{A}{B}`	`\unitvector{1,2}{v}`	⟨1, 2⟩ ⁄ ‖v‖	**Two args:** numerator is auto-wrapped in `\anglevector`; pass the contents to be normed (often the same vector) as the 2nd arg.
`\vec{...}`	`\vec v`	**v**	Redefined to bold (`\mathbf`); overrides the default arrow accent.
`\R`	`\R`	ℝ	Real numbers (`\mathbb{R}`).
`\derivative{y}{x}`	`\derivative{y}{x}`	dy ⁄ dx	Upright `d`.
`\partialderivative{f}{x}`	`\partialderivative{f}{x}`	∂f ⁄ ∂x	
`\grad{f}{P}`	`\grad{f}{x,y}`	∇ f(x, y)	1st arg = function name, 2nd = point/arguments.
`\proj{u}{v}`	`\proj{u}{v}`	proj_**u v**	Both args auto-bolded.
`\comp{u}{v}`	`\comp{u}{v}`	comp_**u v**	Both args auto-bolded.
`\eval{a}{b}`	`F(x)\eval{a}{b}`	F(x) $\vert$ (tall bar, $_a$ lower / ^{b} upper)	Evaluation bar with limits (`\Big
`\jacobian{x,y}{u,v}`	`\jacobian{x,y}{u,v}`	$\vert$ ∂(x,y) ⁄ ∂(u,v) $\vert$	Determinant (`vmatrix`) form of the Jacobian.

4. Structural Macros & Environments

Questions & solutions. Use the paired macros for homework/exam write-ups; `\question` auto-numbers via the `qcount` counter.

```latex
\question{Find the gradient of $f(x,y)=x^2y$.}
\solution{$\grad{f}{x,y} = \anglevector{2xy,\ x^2}$.}
```

→ **Q1.** Find the gradient… / **Solution.** …

Sub-parts. The `parts` environment gives `(a) (b) (c)` labeling with fixed indentation and item spacing:

```latex
\begin{parts}
  \item first part
  \item second part
\end{parts}
```

Prose / essay blocks. Wrap MLA-style written sections in `MLAENV` — Times New Roman (`ptm`) at 12/14.5, double-spaced, with a 0.5in first-line indent. Use it for narrative or reflective passages, not for the math body.

```latex
\begin{MLAENV}
  Body text in Times New Roman, double-spaced...
\end{MLAENV}
```

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5. Theorem Environments

All six are declared, but **no `\theoremstyle` is set**, so every one uses amsthm's default `plain` style (italic body). See §8 to make definitions/remarks upright.

Environment	Numbering	Intended use
`theorem`	per **section** (e.g. Theorem 2.1)	Major results.
`lemma`	sequential, independent	Supporting results.
`corollary`	sequential, independent	Consequences of a theorem.
`proposition`	sequential, independent	Smaller standalone claims.
`definition`	sequential, independent	New terms.
`remark`	sequential, independent	Asides / commentary.

Only `theorem` is tied to the section counter; the rest each carry their own independent counter.

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6. Formatting Conventions

Footnotes — symbols, not numbers. `\renewcommand{\thefootnote}{\fnsymbol{footnote}}` (with `footmisc[symbol]`) prints footnotes as $* \dagger \ddagger \S \P \|$ … Keep ⩽ 9 footnotes per page-reset block, since the symbol set is finite.

Captions. `\captionsetup{labelfont={color=white, bf}, textfont={color=white}}` renders captions **white** — visible only on a dark/colored background. ⚠️ On a white page they vanish; see §8.

Table of contents. `\l@subsection` and `\l@subsubsection` are redefined via `\@tocline` to set TOC indentation (2.5pc / 4pc). Retune spacing by editing those numbers rather than the entries.

Floats. Use `[H]` (from `float`) when a figure must stay exactly in place; fall back to `[ht]` when you want LaTeX to optimize placement.

Links & citations. External links use `\href{URL}{label}`; pair each link with `\cite{key}` so the resource is both clickable and in the bibliography. Ensure every URL includes the `https://` scheme and isn't broken across lines (see §8).

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7. Document Architecture — the Unit Pattern

The body follows one repeating structure. Reuse it verbatim when adding material:

```
\section{Resources:}            % top-level reference list (itemize of \href +
\cite)

\section{Unit NN:}
  \subsection{Lesson Plan:}        % itemize of video/reading links (\href + \cite)
  \subsection{Labs/Homework/Discussions:}
    \paragraph{Homework X.x - X.y:}
    \paragraph{Discussion #n ...:}
    \paragraph{Lab n - (Sec. ...):}

\section{Chapter NN Test:}         % assessment between units
...
\section{Final Exam:}
```

Conventions:

- **One `\section` per unit**, titled `Unit NN:`.
- **Lesson Plan first**, as an `itemize` of `\href`-linked videos/readings, each `\cite`-d.
- **Then** a `Labs/Homework/Discussions` subsection whose items are `\paragraph{...}` headers (Homework ranges, numbered Discussions, numbered Labs).
- **Tests** get their own top-level `\section` between units; the **Final Exam** closes the document.
- The opening figure uses `[H]`; replace the placeholder `\caption{Caption}` and `\label{fig:placeholder}` with real values.

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8. Notes & Things to Watch

A pre-final checklist — several of these are flagged in the preamble's own comments:

- [ ] **Invisible captions.** White caption text disappears on white pages. Either scope the white styling to dark backgrounds only, or drop `color=white` so labels render black/bold.
- [ ] **`\norm` typography.** Doubled single bars (`||...||`) have slightly wrong spacing and can collide. Prefer `\lVert ... \rVert`: `\renewcommand{\norm}[1]{\left\lVert #1 \right\rVert}`
- [ ] **Theorem styles.** With no `\theoremstyle`, definitions and remarks come out italic. Conventionally they're upright — add `\theoremstyle{definition}` before the `definition` declaration, `\theoremstyle{remark}` before `remark`, and `\theoremstyle{plain}` before the theorem/lemma group.
- [ ] **Remove `blindtext`** (and any `\blindtext` / `\Blindtext` calls) before the final build.

- [ ] **Drop redundant packages.** `url` (covered by `hyperref`) and the explicit `caption` line (covered by `subcaption`) can be removed.
- [ ] **Check every `\href` URL.** Some links in the source are missing `https://` or are split across lines — both can break the link. Keep each URL on one unbroken line, with a scheme.
- [ ] **Placeholders.** Fill the title-page figure's caption/label; confirm the printed title (`Math-251 HW and Labs`) is what you want on the cover.
- [ ] **9pt body.** Small for a print write-up — consider 10–11pt unless the small size is intentional.

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9. Quick Reference

```
% --- vectors / notation ---
\anglevector{a,b,c}          ⟨a,b,c⟩
\norm{x}                     ‖x‖
\unitvector{a,b}{v}          ⟨a,b⟩ / ‖v‖
\vec{v}                      bold v
\R                           ℝ

% --- calculus ---
\derivative{y}{x}            dy/dx
\partialderivative{f}{x}     ∂f/∂x
\grad{f}{x,y}                ∇f(x,y)
\proj{u}{v}                  proj_u v   (auto-bold)
\comp{u}{v}                  comp_u v   (auto-bold)
\eval{a}{b}                  |_a^b      (tall bar)
\jacobian{x,y}{u,v}          |∂(x,y)/∂(u,v)|

% --- structure ---
\question{...} / \solution{...}        auto-numbered Q + Solution
\begin{parts} \item ... \end{parts}    (a)(b)(c)
\begin{MLAENV} ... \end{MLAENV}        Times, double-spaced prose

% --- theorems (plain style) ---
theorem (per-section) · lemma · corollary · proposition · definition · remark

% --- load-order rules ---
amsmath/mathtools  ->  amsthm
csquotes           ->  biblatex
(most packages)    ->  hyperref  ->  bibliography
```